

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO3_AT1 — EXPERIMENT

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO3 – Construction (BL3)	Assessment:	Assessment Tool 1 – Experiment
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO3 Statement:	<i>Apply appropriate experimental charting techniques — pie charts, bean plots, violin plots, box plots, and stacked bar charts — to construct accurate visualizations from given data. (BL3)</i>		
Student Name:	K. Amrutha Sandhya	Reg. No.:	192524270
Section / Batch:	AIDS	Date:	11-08-2026

Code of Conduct

I K. Amrutha Sandhya(192524270) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Amrutha

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 75 minutes.
- Graph paper or a ruler is recommended for accurate, to-scale drawing.
- Show all calculations (e.g., pie chart angles, five-number summary, IQR) as part of your answer.
- Every chart must include a title, correctly labeled axes (where applicable), and a legend where relevant.

Annexure A – EXPERIMENT

Rubrics for Calculation **ASSESSMENT RUBRIC**

AT1 — Experiment

CO3 · BT3: Apply ·

This rubric is used to assess student responses to the CO3-AT1 Experiment, in which students are given a small dataset and must construct an accurate pie chart, bean plot, violin plot, box plot, or stacked bar chart.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of the correct chart type (pie, bean, violin, box, or stacked bar) and the elements required to construct it for the given data.	Good understanding of the required chart type and construction elements, with only minor gaps.	Basic understanding of chart construction, but with some misconceptions about the correct chart type or required elements.	Poor understanding of core construction concepts; selects a chart type fundamentally unsuited to the data.
Explanation	25%	Shows detailed, clearly worked calculations (e.g., pie chart angles, five-number summary, IQR, stacked totals) and explains each construction step.	Shows clear calculations and construction steps, with adequate explanation for most parts.	Calculations or construction steps are shown but incomplete, or explanations are superficial.	Little or no working shown; calculations and construction steps are missing or unexplained.
Application	20%	Effectively applies chart construction techniques to produce a complete, accurate chart correctly matched to the data.	Applies construction techniques to produce a correct chart, with only minor errors.	Limited application of construction techniques; the chart partially represents the data but has notable errors.	Fails to apply appropriate construction techniques; the chart does not correctly represent the data.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Organization	15%	The chart is neatly and logically constructed, with a title, correctly labeled axes or slices, a legend where applicable, and elements in a clear order.	The chart is well organized, with only minor issues in labeling, legend, or layout.	Basic construction; required elements (title, labels, legend) are present but poorly arranged or partly missing.	Poorly constructed; the chart is missing key elements such as a title, labels, or a legend.
Accuracy	10%	The chart is completely accurate — all values, proportions, quartiles, and calculations correctly match the given data.	Mostly accurate, with only minor omissions or a small error in one value or calculation.	Partially correct; some plotted values, proportions, or calculations are missing or incorrect.	Largely inaccurate; major errors in plotted values, proportions, or calculations relative to the given data.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Problem 4

Data: A student's 24-hour time allocation: Sleep = 8 hrs, Study = 6 hrs, Leisure = 4 hrs, Other = 6 hrs.

Task: Construct a pie chart for this daily time breakdown, showing the percentage and angle for each activity.

```
# Calculate percentage
percentage <- hours / sum(hours) * 100

# Calculate angle
angle <- hours / sum(hours) * 360

# Display calculations
result <- data.frame(
  Activity = activity,
  Hours = hours,
  Percentage = round(percentge, 2),
  Angle = round(angle, 2)
)

print(result)

# Pie chart
pie(hours,
  labels = paste(activity, round(percentge, 2), "%"),
  main = "Student's 24-Hour Time Allocation")
```

(Top Level) R Script

Background Jobs x

R4.6.1 · ~/

```
25
28
30
35
60

plot(data, aes(x = "", y = Delivery_Time)) +
  geom_violin(fill = "skyblue") +
  labs(
    title = "Delivery Times",
    x = "Orders",
    y = "Delivery Time (minutes)"
  ) +
  theme_minimal()
```

Global Environment

delivery	num [1:10]	15 18 20 22 25 25 28 30 35 60
frequency	num [1:4]	8 5 4 3
fruit	chr [1:4]	"Apple" "Banana" "Orange" "Grap...
hours	num [1:4]	8 6 4 6
IQR_value		16.5
labels	chr [1:7]	"Mon 13 %" "Tue 14 %" "wed 15 %...
lower	Named num	-3.75
marks	num [1:9]	40 50 55 60 60 60 65 70 80
outliers		90

Files Plots Packages Help Viewer Presentation

Zoom Export Publish

Student's 24-Hour Time Allocation

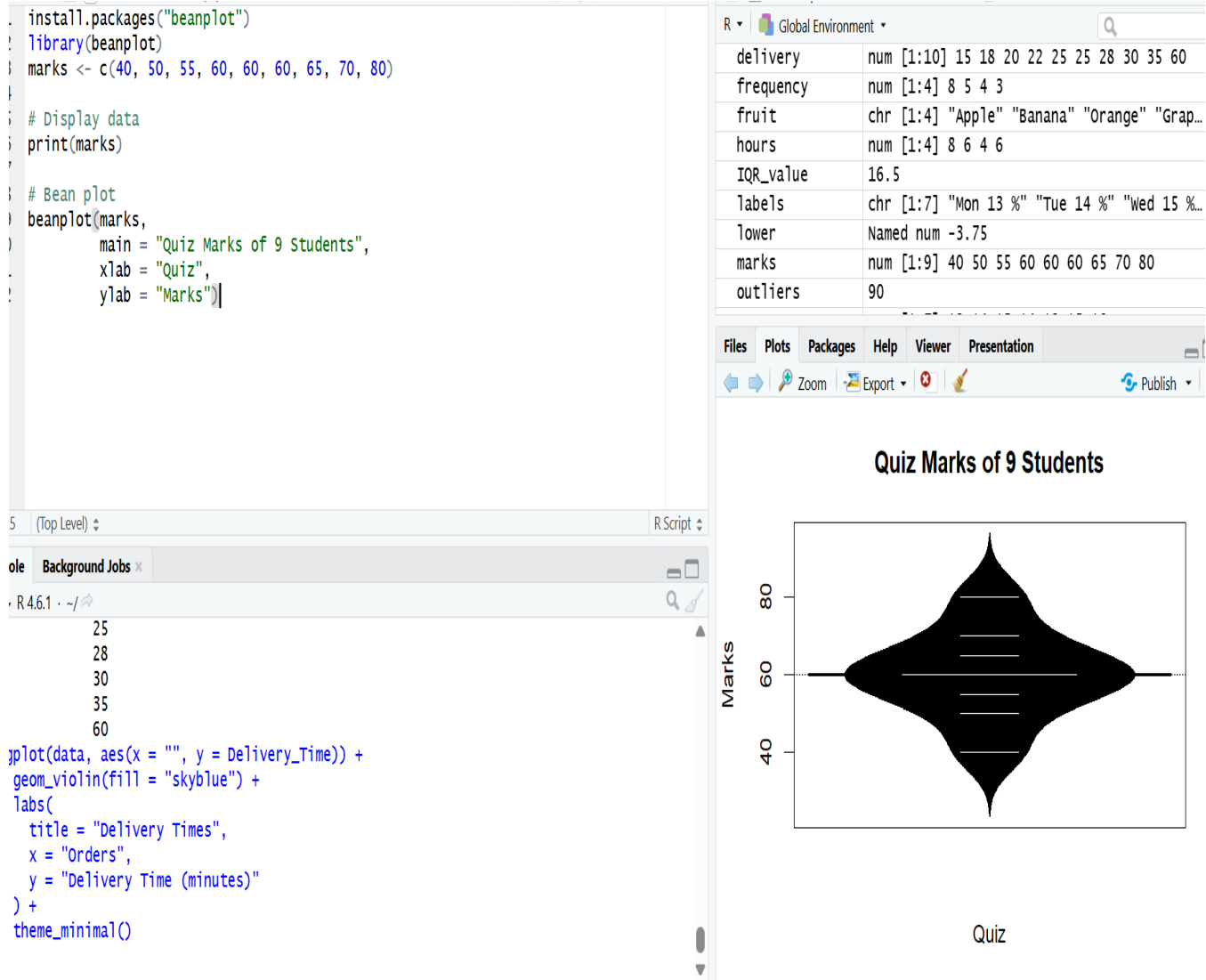
A pie chart titled "Student's 24-Hour Time Allocation" showing the distribution of a student's 24-hour day. The chart is divided into four segments: Sleep (33.33%, white), Study (25%, blue), Leisure (16.67%, red), and Other (25%, cyan). The segments are labeled with their respective activity names and percentages.

Activity	Hours	Percentage
Sleep	8	33.33 %
Study	6	25 %
Leisure	4	16.67 %
Other	6	25 %

Problem 6

Data: Quiz marks of 9 students: 40, 50, 55, 60, 60, 60, 65, 70, 80.

Task: Construct a bean plot for this data and identify any repeated values that become visible in the plot.



Problem 13

Data: Delivery times (in minutes) for 10 orders: 15, 18, 20, 22, 25, 25, 28, 30, 35, 60.

Task: Construct a violin plot and identify where most delivery times are concentrated.

```
install.packages("ggplot2")
library(ggplot2)
delivery <- c(15, 18, 20, 22, 25, 25, 28, 30, 35, 60)
```

```
# Create data frame
data <- data.frame(
  Delivery_Time = delivery
)

# Display data
print(data)
ggplot(data, aes(x = "", y = Delivery_Time)) +
  geom_violin(fill = "skyblue") +
  labs(
    title = "Delivery Times",
    x = "Orders",
    y = "Delivery Time (minutes)"
  ) +
  theme_minimal()
```

(Top Level) ▾

R Script ▾

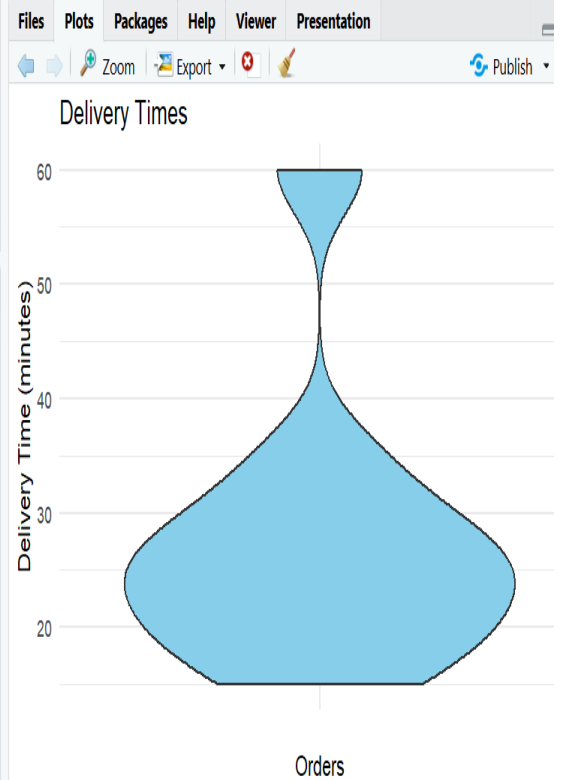
Background Jobs x

R 4.6.1 · ~/

25
28
30
35
60

```
plot(data, aes(x = "", y = Delivery_Time)) +
  geom_violin(fill = "skyblue") +
  labs(
    title = "Delivery Times",
    x = "Orders",
    y = "Delivery Time (minutes)"
  ) +
  theme_minimal()
```

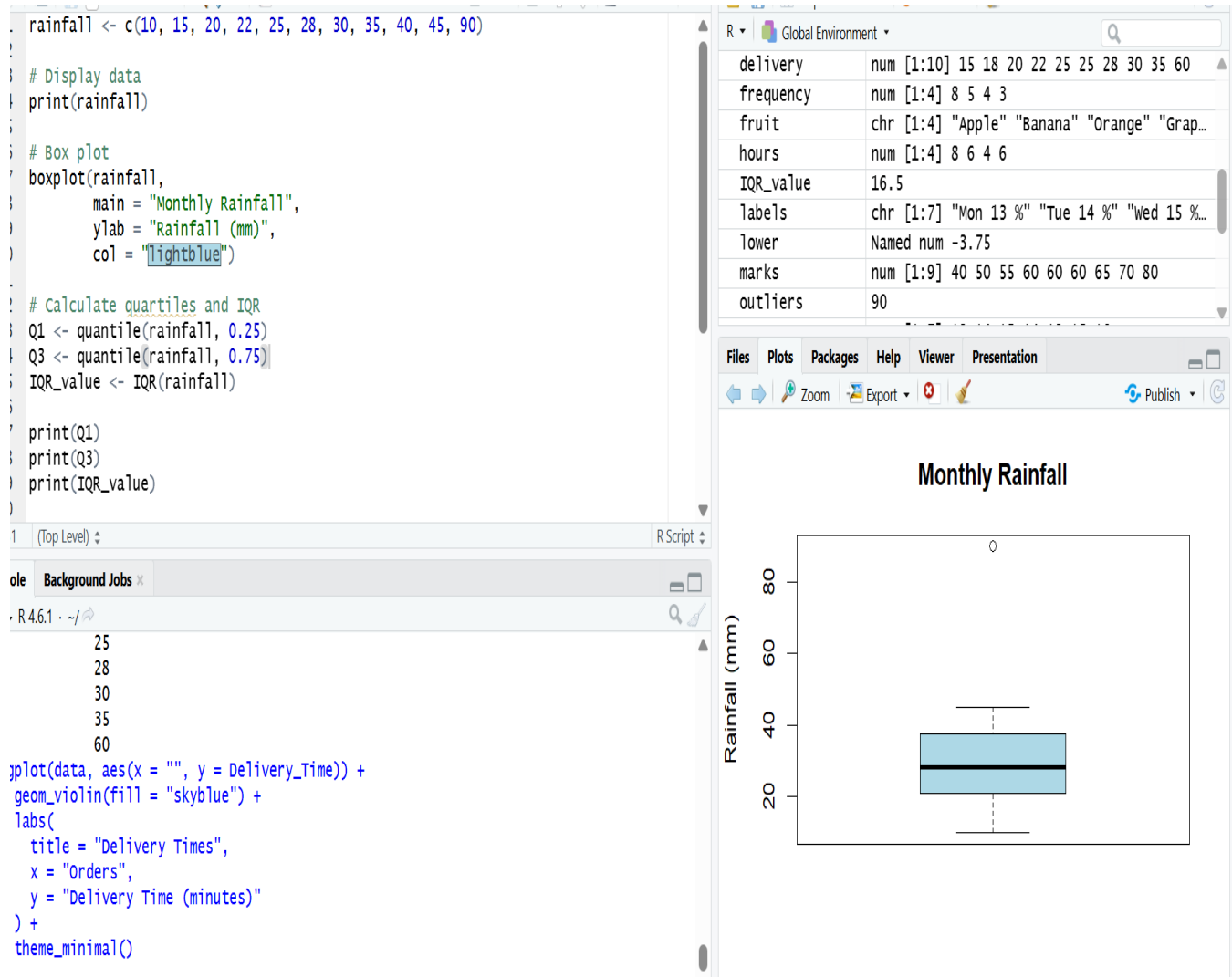
delivery	num	[1:10]	15	18	20	22	25	25	28	30	35	60
frequency	num	[1:4]	8	5	4	3						
fruit	chr	[1:4]	"Apple"	"Banana"	"Orange"	"Grap.						
hours	num	[1:4]	8	6	4	6						
IQR_value			16.5									
labels	chr	[1:7]	"Mon 13 %"	"Tue 14 %"	"Wed 15 %"							
lower	Named num		-3.75									
marks	num	[1:9]	40	50	55	60	60	60	65	70	80	
outliers			90									



Problem 20

Data: Monthly rainfall (in mm) over 11 months: 10, 15, 20, 22, 25, 28, 30, 35, 40, 45, 90.

Task: Construct a box plot and identify which month's rainfall value is an outlier.



Problem 22

Data: Quarterly sales by product: Q1 — Product A = 100, Product B = 80. Q2 — Product A = 120, Product B = 90.

Task: Construct a stacked bar chart comparing total sales and product composition across both quarters.

